CO1 - AT1

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Course :- CSA-0414 - Operating System

1) a) A modern system supporting multiple users and processes is best suited by :-

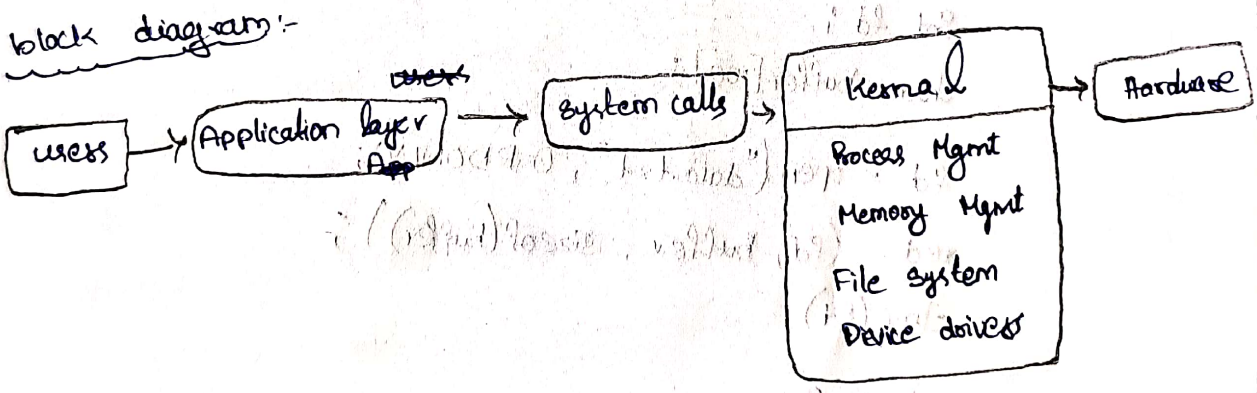
1. Multi-user Operator System :-
   - allows multiple users to access the system simultaneously.
   - Ex :- Linux, UNIX.

2. Multiprogramming Operator System :-
   - keeps several programs in memory and executes them efficiently
   - Maximizes CPU utilization.

3. Time-Sharing Operating System :-
   - Allocates CPU time slices to users and processes.

Overall, A combination of Multi-user and Time-Sharing OS is most suitable because it enables concurrent user access while efficiently sharing CPU, memory, and I/O resources.

b) block diagram :-



c) System Calls :-

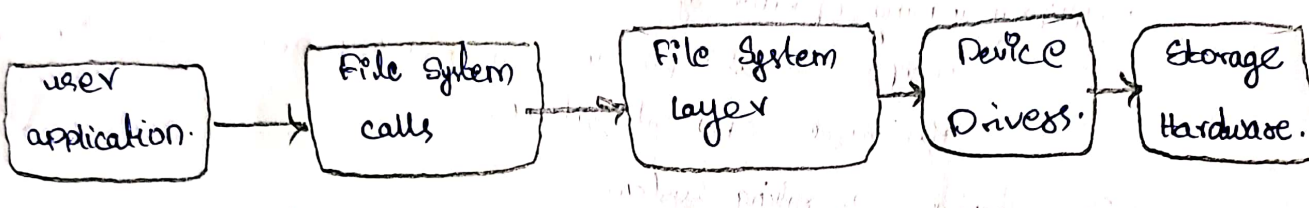
(i) fork() - creates a child process
(ii) exec() - Loads and executes a new program.
(iii) wait() - Parent waits for childs completion.
(iv) exit() - Terminates process.

2/ a)

- File :- (i) open()
  (ii) read()
  (iii) write()
  (iv) close()

- Device :- (i) read()
  (ii) write()
  (iii) ioctl()
  (iv) close().

b) Layered Structure for File Operations



c) C-based System Call sequence :-

```
#include <fcntl.h>
#include <unistd.h>
int main() {
    int fd;
    char buffer[100];
    fd = open("data.txt", O_RDONLY);
    red = (fd, buffer, sizeof(buffer));
    close(fd)

    return 0;
}
```

Here, open() → opens the file.
read() → reads the file
close() → closes the file.

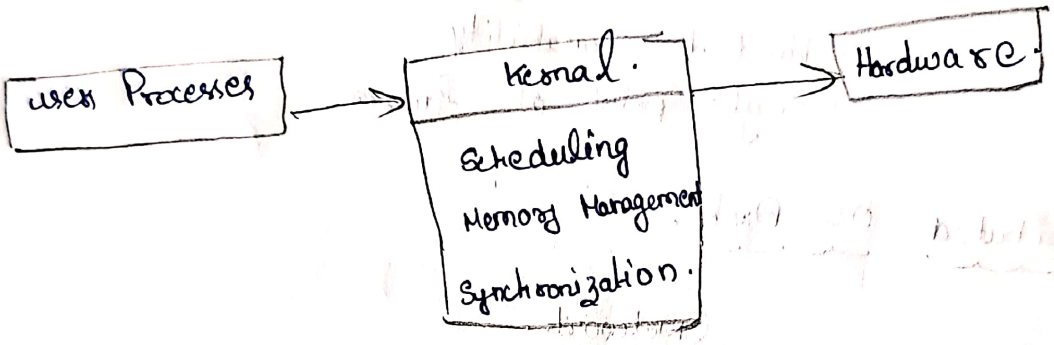
b) The suitable OS is a Real-Time Operating system (RTOS)

because, (i) Provides deterministic response time.

(ii) Ensure critical tasks meet deadlines

(iii) Suitable for Industrial automation and embedded syste

c) Interaction b/w Hardware, Kernal, and User Processes :-



d) System calls for process Synchronization :-

(i) fork() → creats cooperating processes.

(ii) wait() → waits for another process

(iii) sem_wait() → Acquires semaphore.

(iv) sem_Post() → Releases semaphore.

These, calls prevent race conditions and ensure correct process coordination.

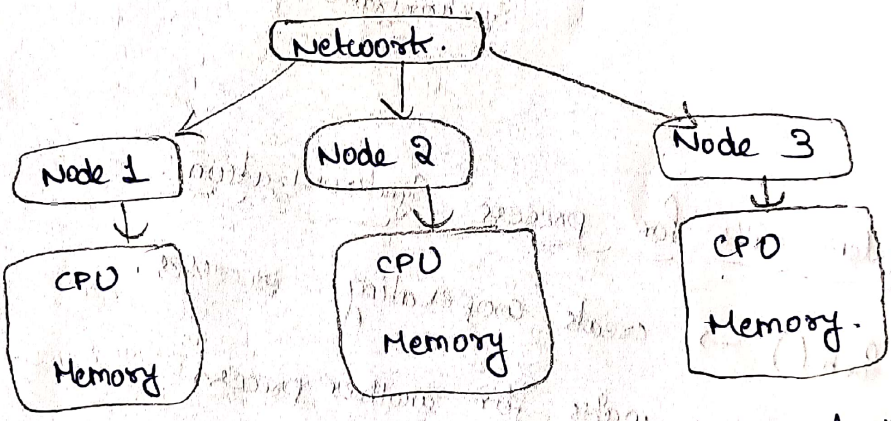
4) a)

The system uses a Distributed Operating System.

(i) Distributed :-
- Multiple, interconnected computers
- Resources shared across network.
- Better scalability.
- High reliability.

(ii) Centralized :-
- Single computer system.
- Resources managed locally
- Limited scalability.
- Single point of failure.

b) Distributed OS Arch :-



- All nodes communication and share resources through the networ

c) Sequence of System calls :-

Ex:

```
pipe(fd);                          // (i) pipe()
write(fd[1], "Hello", 5);          // (ii) write()
read(fd[0], buffer, 5);            // (iii) read()
close(fd[0]);                      // (iv) close()
close(fd[1]);
```

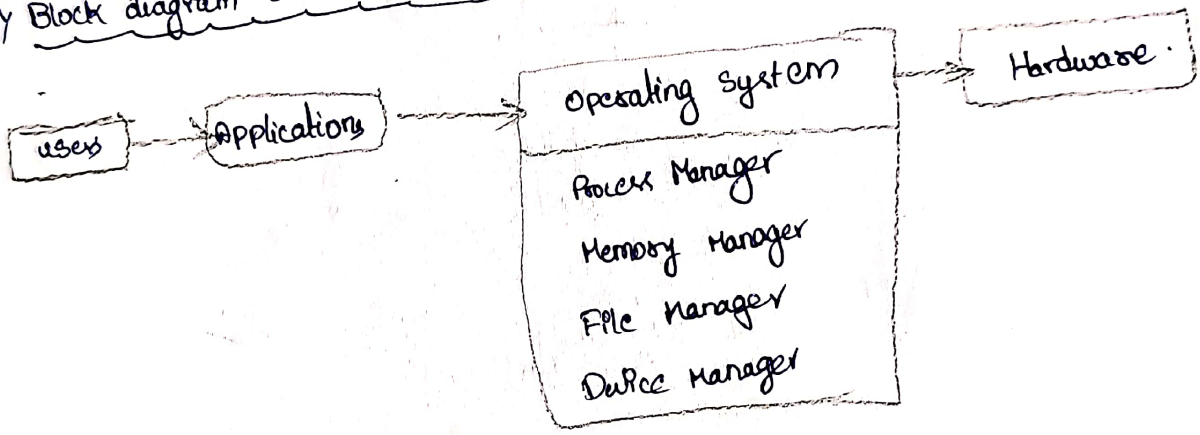
The three major functions of Operating System are process Management, Memory Management, I/O Management.

(i) Process Scheduling :- • Allocates CPU to processes
• Performs context switching.
• Ensures fair execution.

(ii) Memory Allocation : • Allocates and deallocates memory.
• Maintains Virtual Memory

(iii) Device Management : • controls I/O devices.

b) Block diagram of OS components:



c) System calls example :

(i) Memory allocation :
- brk()
- sbrk()
- mmap()

(ii) Device Management :
- open()
- read()
- write()
- ioctl()
- close().

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